

Day 6 - Swap Two Numbers

Question

Do numbers diye gaye hain.

Unki values swap karni hain.

Example

Input

```
a = 10  
b = 20
```

Output

```
a = 20  
b = 10
```

Approach 1 - Using Third Variable (Best for Interviews)

Logic

Ek temporary variable me value store karo.

Dry Run

```
a = 10
b = 20

temp = a → 10
a = b → 20
b = temp → 10
```

Java Code

```
public class Day6 {
    public static void main(String[] args) {

        int a = 10;
        int b = 20;

        int temp = a;
        a = b;
        b = temp;

        System.out.println("a = " + a);
        System.out.println("b = " + b);
    }
}
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Approach 2 - Without Third Variable (Addition/Subtraction)

Logic

$a = a + b$

$b = a - b$

$a = a - b$

Dry Run

$a = 10$

$b = 20$

$a = 30$

$b = 30 - 20 = 10$

$a = 30 - 10 = 20$

Java Code

```
public class Day6 {  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 20;  
  
        a = a + b;  
        b = a - b;  
        a = a - b;  
  
        System.out.println(a);  
        System.out.println(b);  
    }  
}
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Problem

Large numbers par overflow ho sakta hai.

```
a = 2147483647;  
b = 1;
```

Isliye production code me avoid karte hain.

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Approach 3 - Multiplication / Division

Logic

```
a = a * b  
b = a / b  
a = a / b
```

Example

```
a = 10  
b = 20  
  
a = 200  
  
b = 200/20 = 10  
  
a = 200/10 = 20
```

Java Code

```
public class Day6 {  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 20;  
  
        a = a * b;  
        b = a / b;  
        a = a / b;  
  
        System.out.println(a);  
        System.out.println(b);  
    }  
}
```

Problem

Agar koi value 0 ho to fail ho jayega. Division by Zero

Approach 4 - XOR Bitwise Operator ☐

Interview Favorite

Logic

$a = a \oplus b$

$b = a \oplus b$

$a = a \oplus b$

Dry Run

$a = 10$

$b = 20$

$a = 10 \oplus 20$

$b = (10 \oplus 20) \oplus 20 = 10$

$a = (10 \oplus 20) \oplus 10 = 20$

Java Code

```
public class Day6 {  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 20;  
  
        a = a ^ b;  
        b = a ^ b;  
        a = a ^ b;  
  
        System.out.println(a);  
        System.out.println(b);  
    }  
}
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Interview Follow-Up Questions

Q1. Best Approach?

Interview me:

```
int temp = a;  
a = b;  
b = temp;
```

Batana chahiye.

Readable aur safe hai.

Q2. XOR Approach kyu famous hai?

Extra variable use nahi hota.

Bit manipulation ka knowledge show hota hai.

Q3. Addition/Subtraction Method ka drawback?

Overflow.

```
Integer.MAX_VALUE
```

ke case me wrong answer aa sakta hai.

Real Interview Discussion

Interviewer:

Swap two numbers without third variable.

Candidate:

```
a = a + b;  
b = a - b;  
a = a - b;
```

Interviewer:

Any problem?

Candidate:

Yes Sir,
Integer Overflow ho sakta hai.

Fir XOR Approach explain karo.

Ye strong answer mana jata hai.

Summary

Approach	Time	Space	Recommended
Third Variable	O(1)	O(1)	❑ Best
Addition/Subtraction	O(1)	O(1)	Good
Multiplication/Division	O(1)	O(1)	Not Preferred
XOR	O(1)	O(1)	Interview Favorite

Practice Questions

Swap:

1

```
a = 5;  
b = 10;
```

2

```
a = 100;  
b = 200;
```

3

```
a = -5;  
b = 10;
```

4

```
a = 0;  
b = 50;
```

5

```
a = 999;  
b = 111;
```

Predict final values.